🔐 Hybrid VPN – In-Depth Theory

🔷 1. What is a Hybrid VPN?

A **Hybrid VPN** combines multiple VPN technologies and tunneling strategies (like SSL, IPsec, MPLS, GRE, etc.) to create **secure, flexible, and scalable** VPN solutions.

🔸 Use Cases:

- SSL VPN for remote users + IPsec for site-to-site
- GRE tunnel inside an IPsec tunnel for dynamic routing
- MPLS core with IPsec at the edge for encryption

📌 **Example:**
A company may use IPsec for permanent inter-office connections and OpenVPN (SSL-based) for remote workers. Together, this forms a **Hybrid VPN architecture**.

---

2. IPsec (Internet Protocol Security)

IPsec is a framework of protocols that operate at the **network layer** to secure IP communications by authenticating and encrypting each IP packet.

Components:

- **AH (Authentication Header)** – Provides integrity and authentication
- **ESP (Encapsulating Security Payload)** – Provides confidentiality, integrity, and optional authentication
- **IKE (Internet Key Exchange)** – Manages keys and security associations (SA)

Modes:

- **Transport Mode**

    - Only encrypts payload
    - Used in **host-to-host** connections

- **Tunnel Mode**
    - Encrypts the entire IP packet and adds a new header
    - Used in **gateway-to-gateway** (site-to-site) or **remote access VPNs**

---

3. Tunnel Mode vs Transport Mode

Feature	Tunnel Mode	Transport Mode
What is encrypted?	Entire IP packet	Only payload and upper layers
Use case	Site-to-site VPN, Remote Access	End-to-end host communications
Header change	New IP header added	Retains original IP header
Protocol used	ESP or AH	ESP or AH

---

4. IPv6 VPN

With the expansion of IPv6 networks, traditional VPN protocols needed updates:

IPv6-Compatible VPN Protocols:

- **IPsec with IPv6** – Fully supported
- **OpenVPN** – Supports IPv6 inside tunnel and IPv6 VPN servers

- **WireGuard** – Fully supports IPv6 endpoints and routing

Benefits:

- No NAT (simplifies VPN routing)
- Larger address space
- End-to-end communication is restored

---

5. Split Tunnel vs Full Tunnel VPN

Split Tunnel VPN

-

Only specific traffic is routed through the VPN tunnel (e.g., internal resources)

- All other traffic (e.g., web browsing) uses the regular internet connection

Pros: Saves bandwidth, improves speed
Cons: Security risk—external traffic is not encrypted

Full Tunnel VPN

- **All traffic** is routed through the VPN tunnel (internal + external)
- More secure, especially on untrusted networks like public Wi-Fi

Pros: Maximum security
Cons: Can increase latency, load on VPN server

📊 Comparison Table

Feature	Split Tunnel	Full Tunnel
Traffic routed	Only internal traffic	All traffic (internal + external)
Speed	Faster	Slower (all through VPN)
Security	Less secure	More secure
Use case	Access intranet + local browsing	Secure remote access